

Forms in Angular

Template-Driven Form

- ✓ Simple Basic Form
- ✓ Easy to start with
- ✓ Based on template (HTML)

Reactive Form

- ✓ Complex form with more control
- ✓ Structure of form is defined in TypeScript class



Reactive Form in Angular

- Reactive forms are forms where we define the structure of the form in the component class. i.e., we create the form model with Form Groups, Form Controls, and Form Arrays.
- We also define the validation rules in the component class.
- Then, we bind it to the HTML form in the template.
- This is different from the template-driven forms, where we define the logic and controls in the HTML template.



Route Parameter

The route parameters are the dynamic part of the route whose value can change. These parameters provides a way to pass extra information to a given route.

`localhost:4200/Books/Author/Stefen-King`

`localhost:4200/Books/Author/Sarah-Williams`

`localhost:4200/Books/Author/John-Smith`



Route Parameter

The route parameters are the dynamic part of the route whose value can change. These parameters provides a way to pass extra information to a given route.

`localhost:4200/Books/:id`

`localhost:4200/Books/Author/:author`





EXPLORER

ANGULAR-ROUTING

- > contact
- > courses
 - > course-detail
 - # course-detail.component.css
 - <> course-detail.component.html
 - TS course-detail.component.ts
 - # courses.component.css
 - <> courses.component.html
 - TS courses.component.ts
- > footer
- > header
- > home
- > login
- > Models
- > not-found
- > Services
- # app.component.css
- <> app.component.html

> OUTLINE

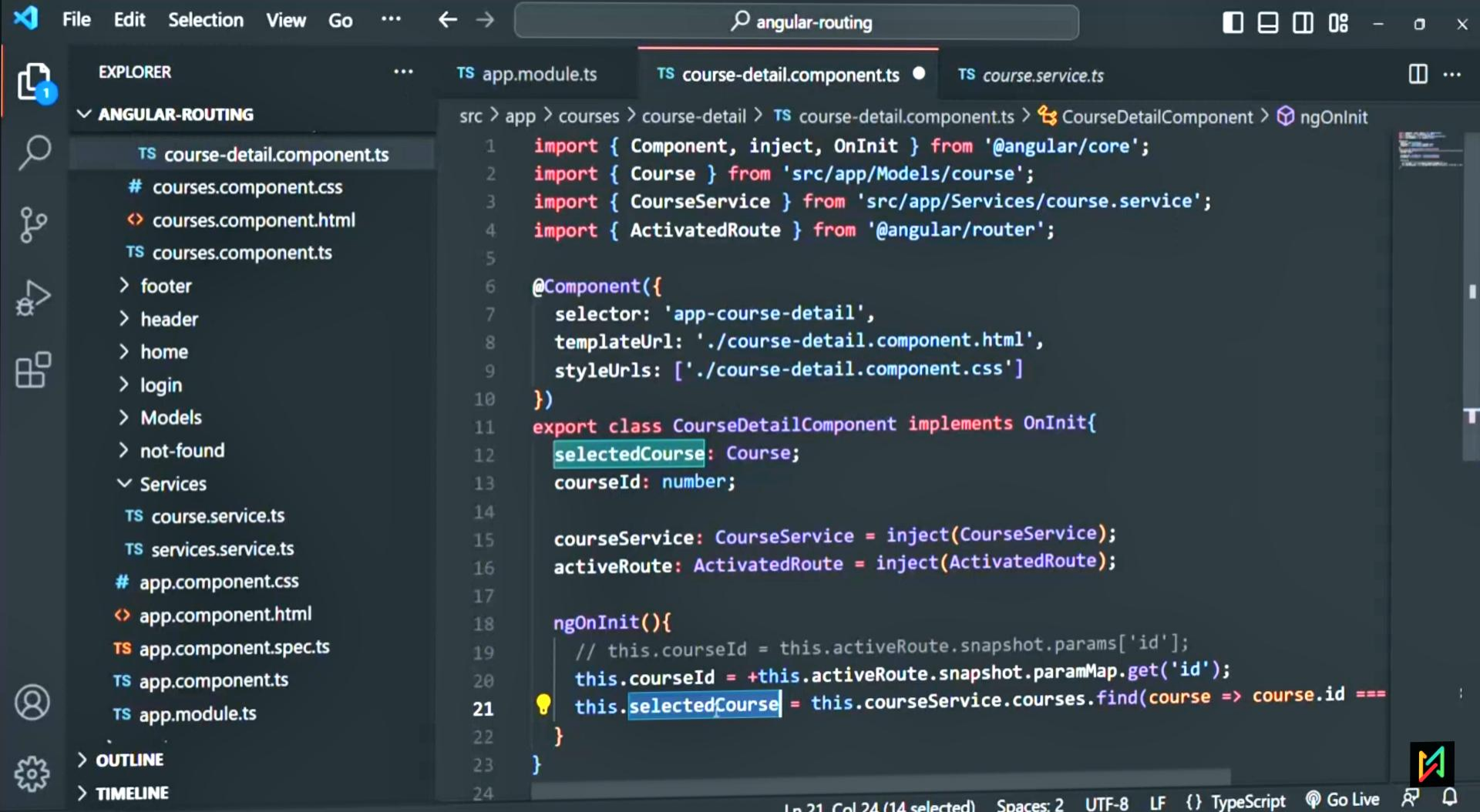
> TIMELINE

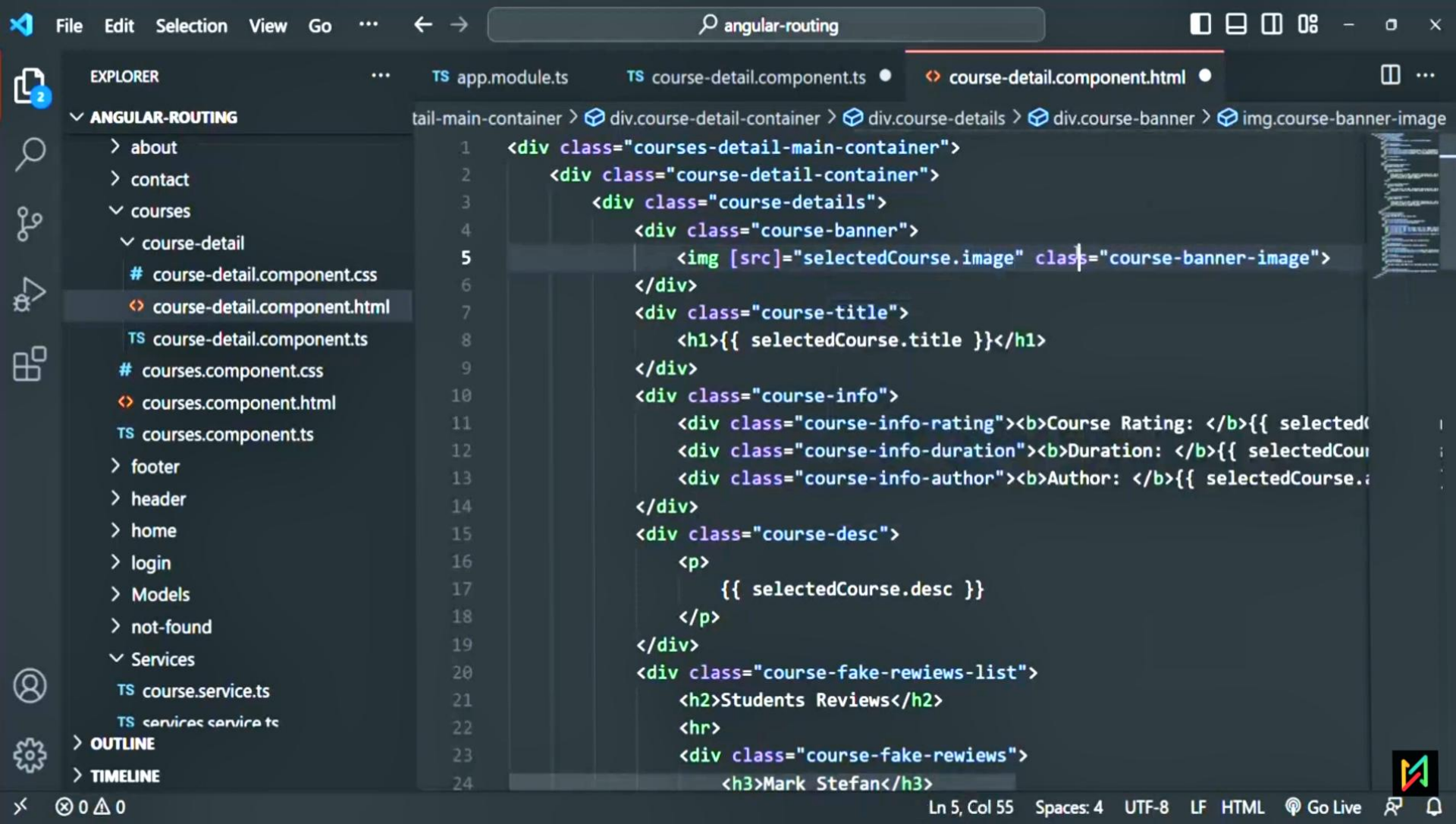
TS app.module.ts

TS course-detail.component.ts

```
src > app > courses > course-detail > TS course-detail.component.ts > CourseDetailComponent > ngOnInit

1  import { Component, inject, OnInit } from '@angular/core';
2  import { Course } from 'src/app/Models/course';
3  import { CourseService } from 'src/app/Services/course.service';
4  import { ActivatedRoute } from '@angular/router';
5
6  @Component({
7    selector: 'app-course-detail',
8    templateUrl: './course-detail.component.html',
9    styleUrls: ['./course-detail.component.css']
10 })
11 export class CourseDetailComponent implements OnInit{
12   selectedCourse: Course;
13   courseId: number;
14
15   courseService: CourseService = inject(CourseService);
16   activeRoute: ActivatedRoute = inject(ActivatedRoute);
17
18   ngOnInit(){
19     this.courseId = this.activeRoute.snapshot.params['id'];
20   }
21 }
22
```





FileEditSelectionViewGo...<=>angular-routing

EXPLORER

ANGULAR-ROUTING

> about

> contact

> courses

> course-detail

course-detail.component.css

<> course-detail.component.htmlTS course-detail.component.ts# courses.component.css

<> courses.component.htmlTS courses.component.ts

> footer

> header

> home

> login

> Models

> not-found

> Services

TS course.service.tsTS services.service.ts

OUTLINE

TIMELINE

course-detail.component.ts

course-detail.component.html

courses.component.html

iv.courses-page-all-courses > div.courses-page-course-list > div.course-card > div.course-action-buttons > button.btn

18<div class="course-desc">

19 {{ course.desc.substring(0, 120) }}

20</div>

21<div class="course-price-rating">

22 <div class="course-price">PRICE: {{ course.price | curren

23 <div class="course-rating">RATINGS: {{ course.rating }}</

24</div>

25<div class="course-action-buttons">

26 <button class="btn">BUY NOW</button>

27 <button class="btn" [routerLink]="'/Courses/Course/' + course.id"

28</div>

29</div>

30</div>

31</div>

32</div>

33

Ln 27, Col 57

Spaces: 4

UTF-8

LF

HTML

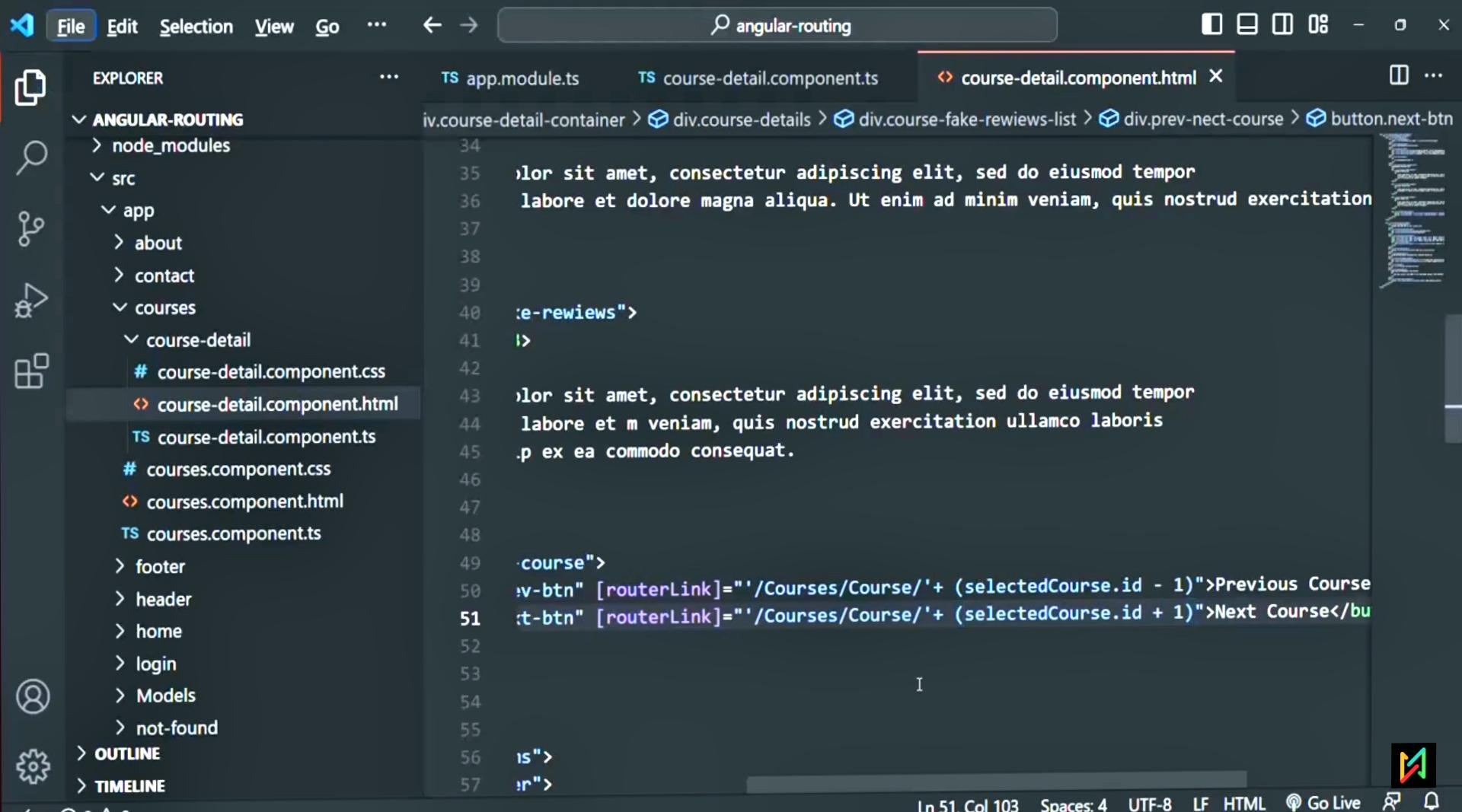
Go Live

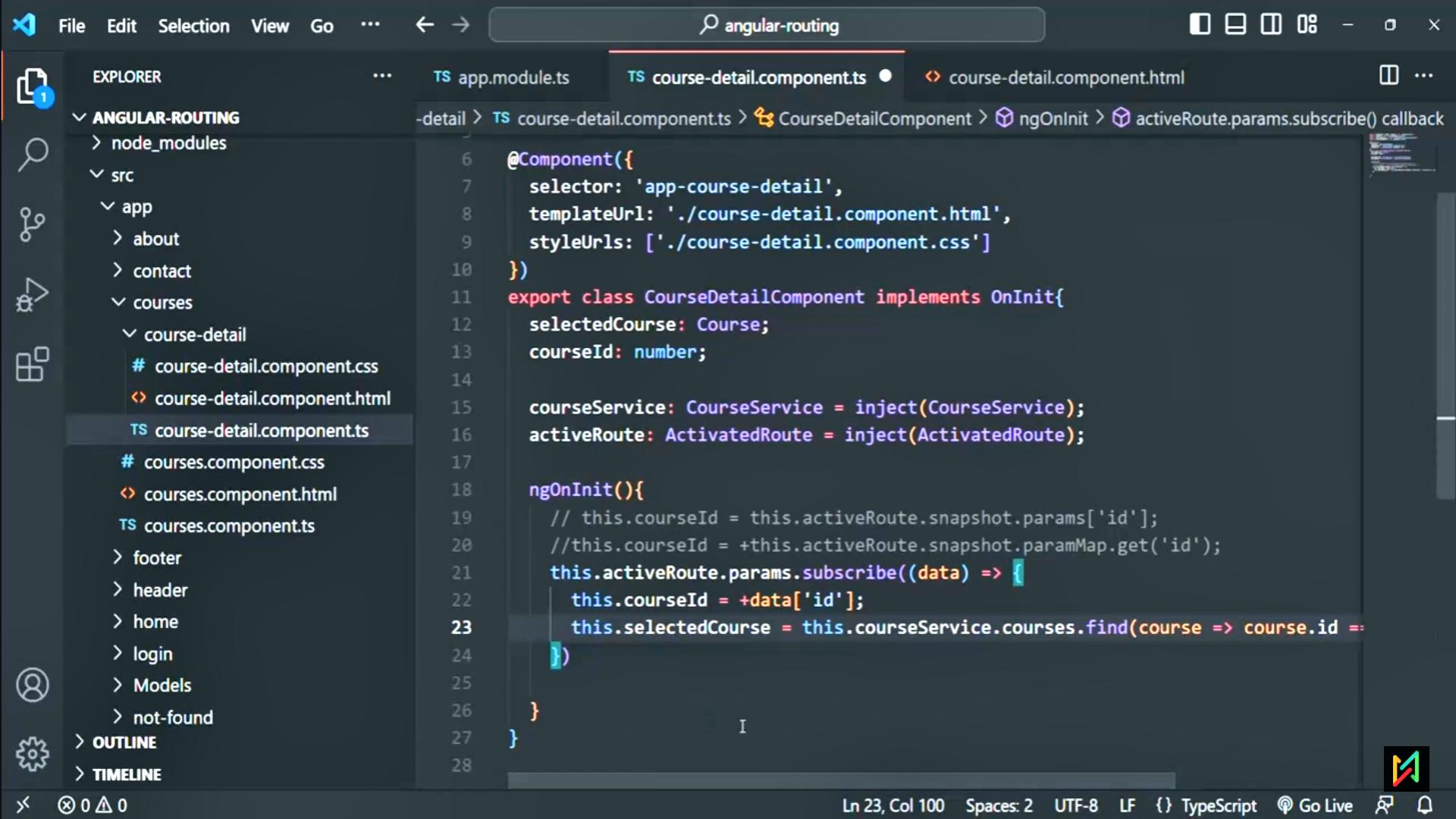
Snapshot property

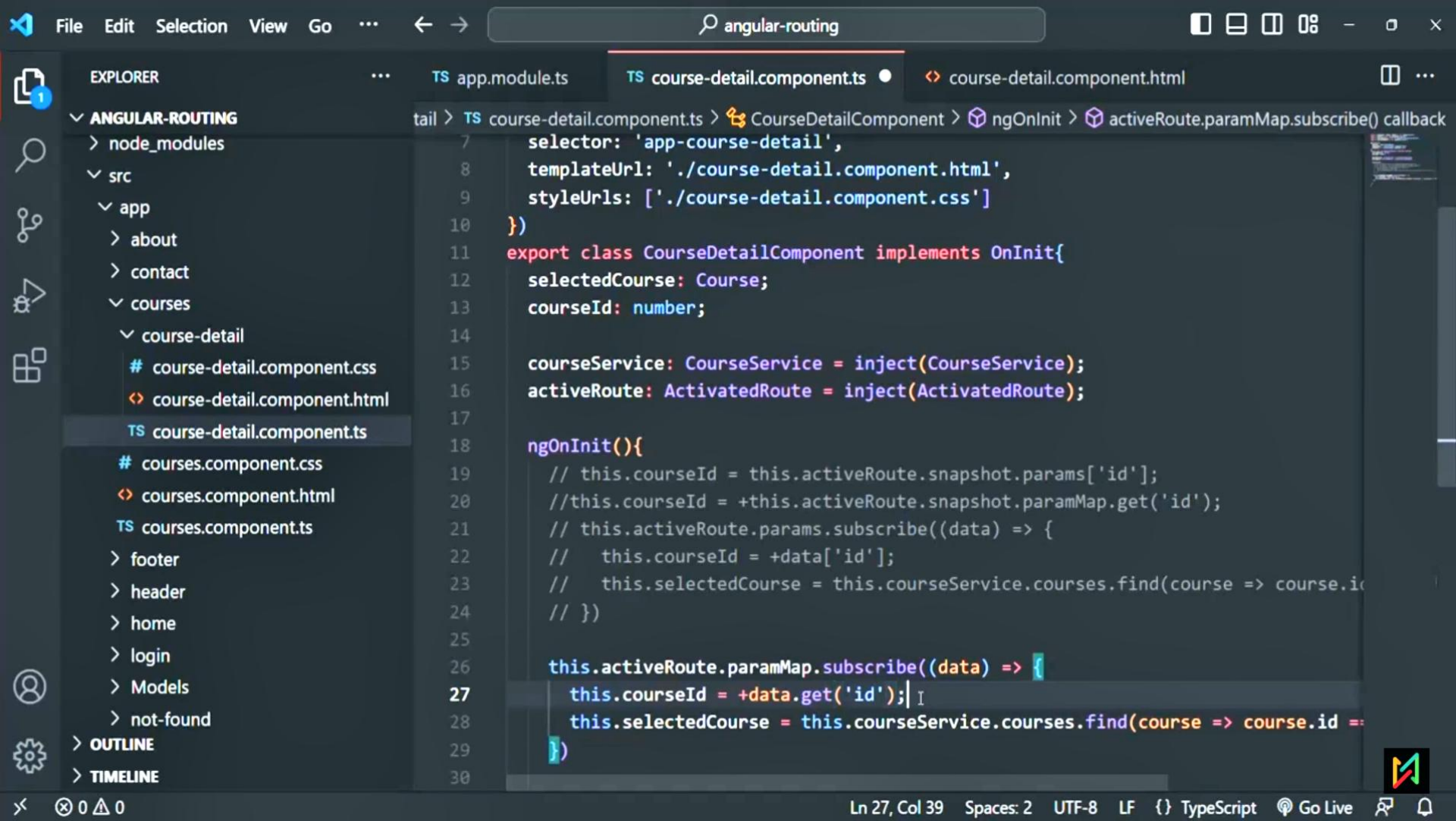
The snapshot property contains the current value of the route. If the value of the route parameter changes after we have retrieved the value of route, that change will not get captured.

```
this.activeRoute.snapshot paramMap.get('id');
```





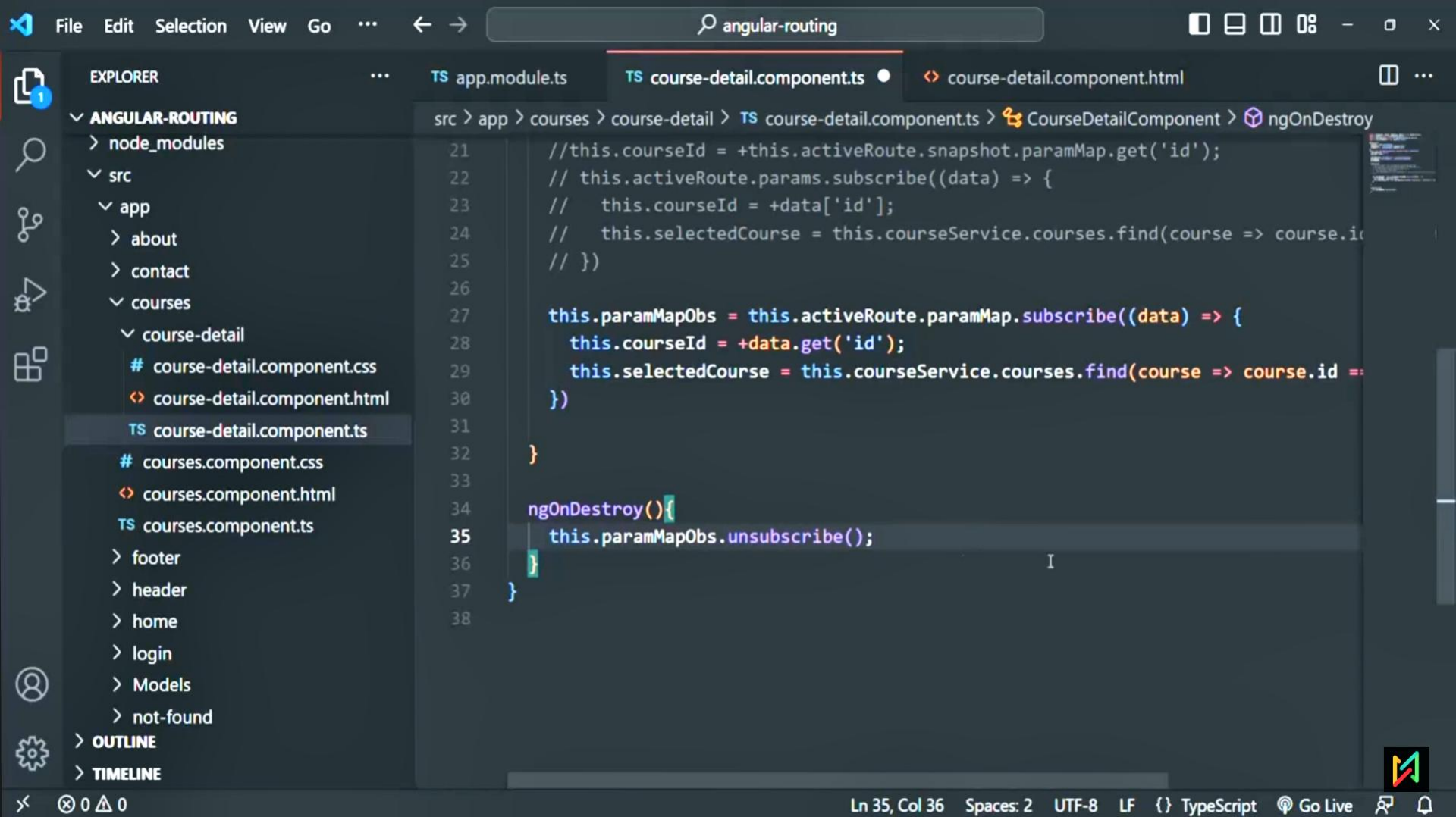




The paramMap Observable

By subscribing to the paramMap observable (or to params observable), you will get a notification when the value changes. Hence, you can retrieve the latest value of the parameter and update the component accordingly.





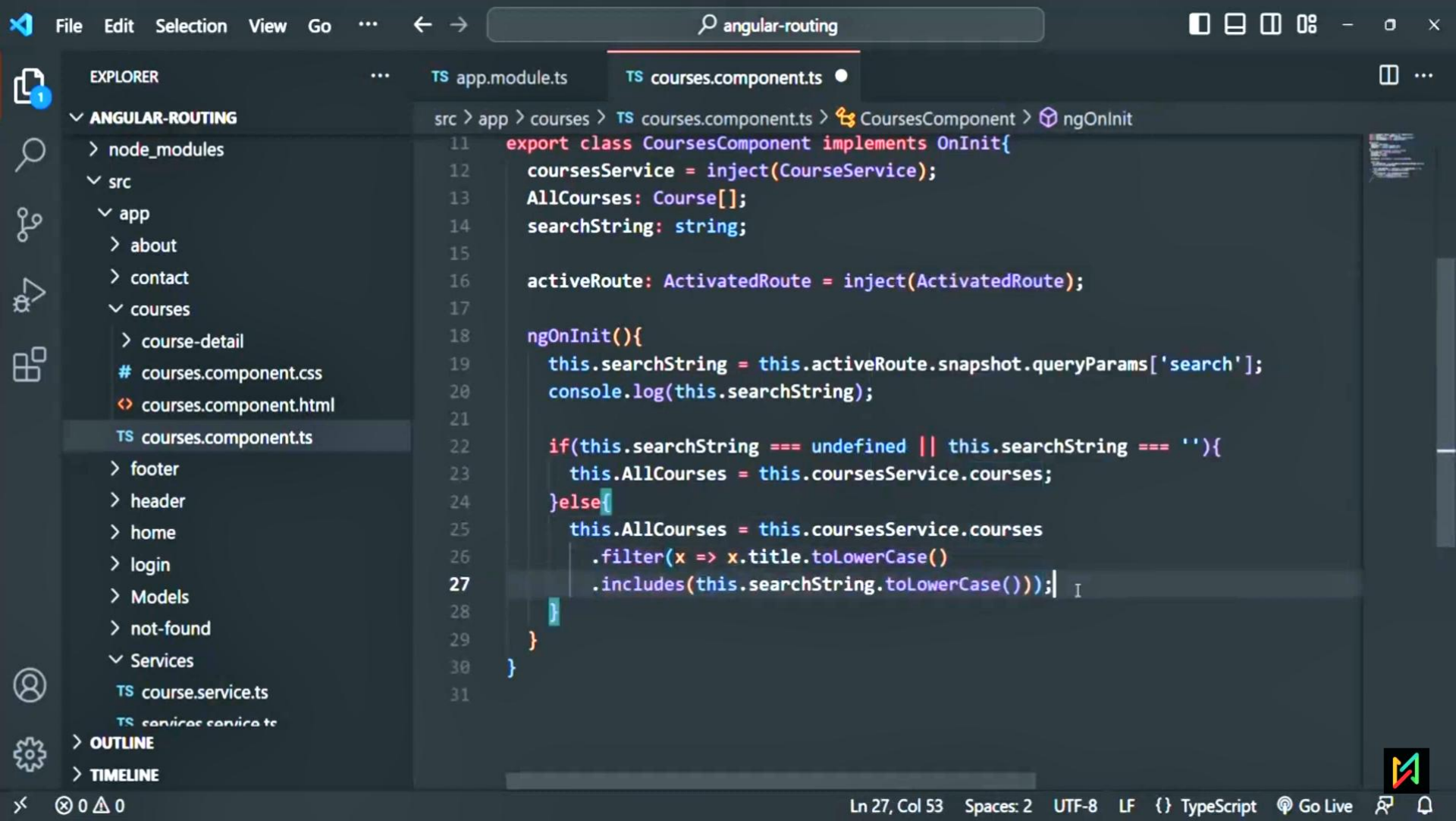
What is Query String

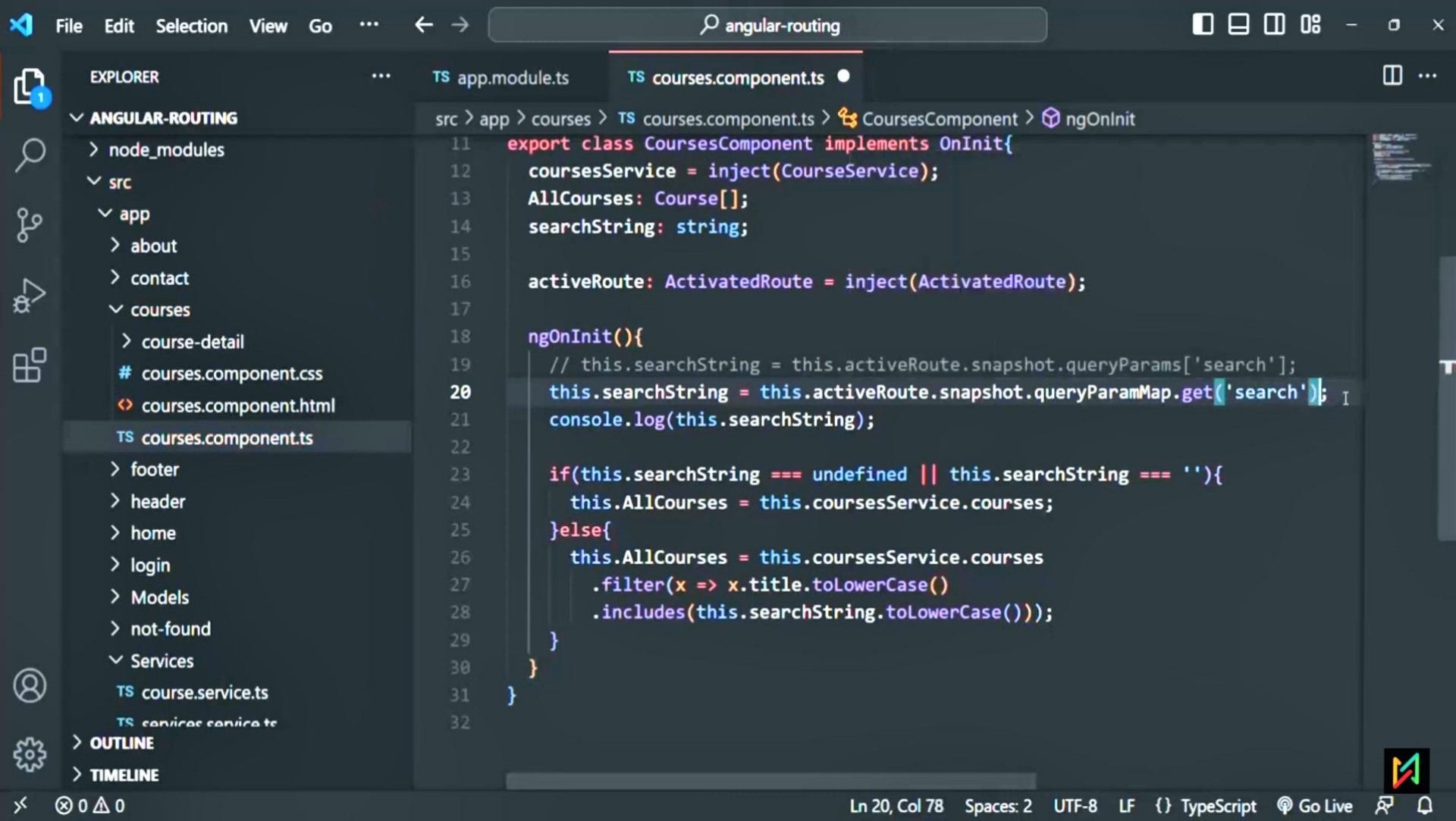
Query Strings are the optional data that we can pass to a component through a route. These query strings are added at the end of the route after a `?`

`localhost:4200/Books/Author?name=stefen-king`

`localhost:4200/Products/product?id=12345&name=iphone`







EXPLORER

TS app.module.ts

TS courses.component.ts

ANGULAR-ROUTING

> node_modules

> src

> app

> about

> contact

> courses

> course-detail

courses.component.css

<> courses.component.html

TS courses.component.ts

> footer

> header

> home

> login

> Models

> not-found

> Services

TS course.service.ts

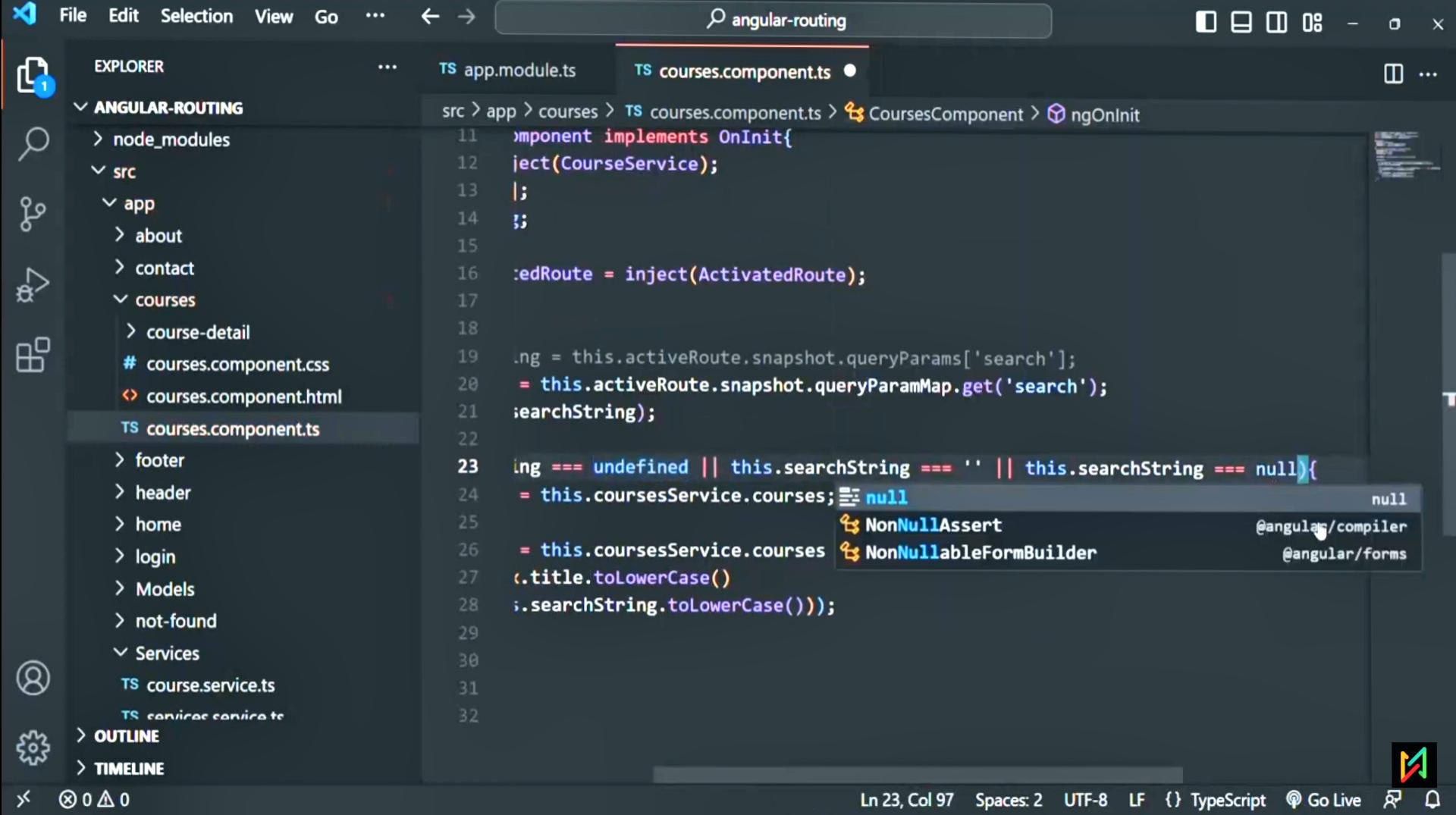
TS courses.service.ts

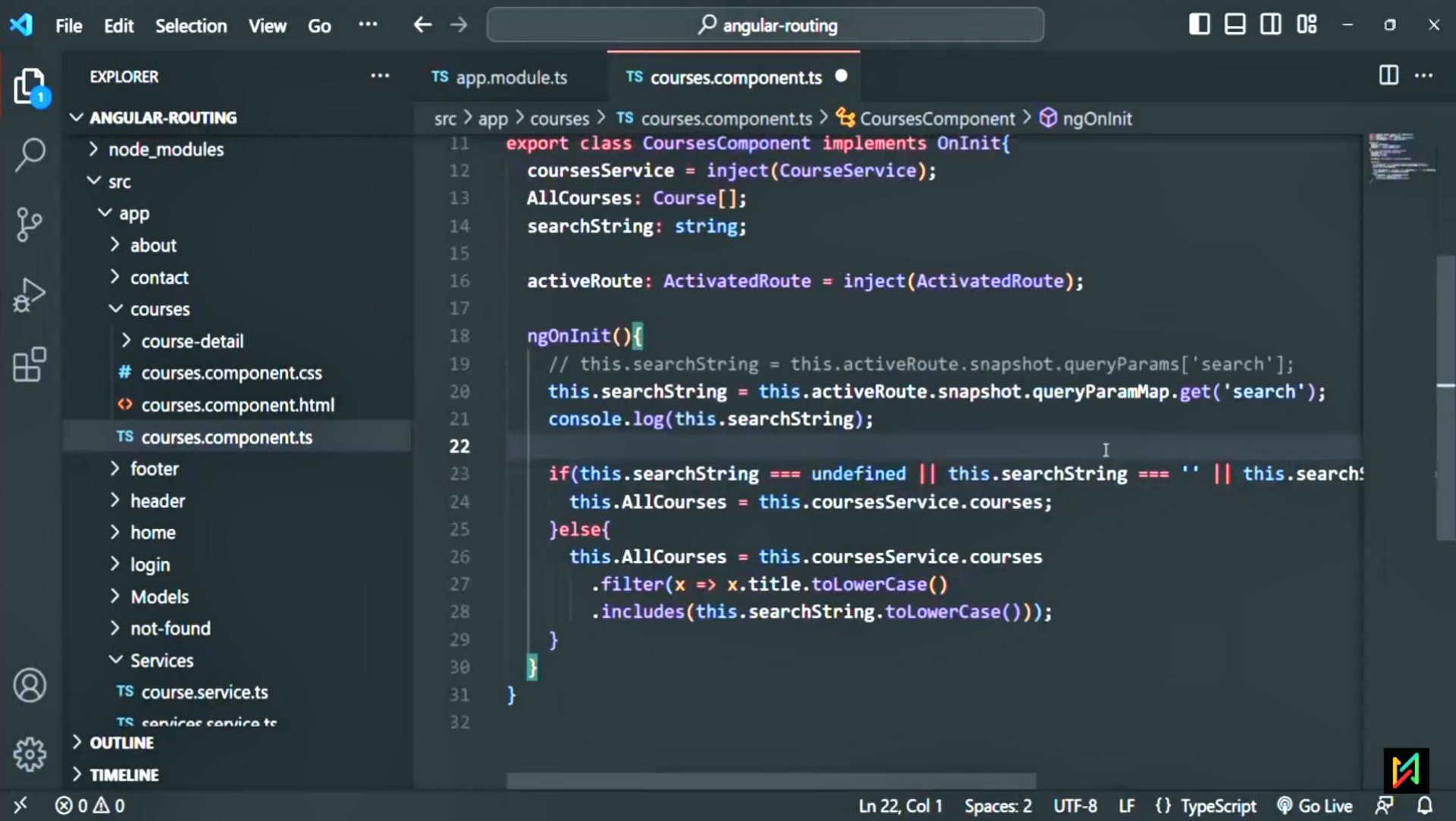
> OUTLINE

> TIMELINE

src > app > courses > TS courses.component.ts > CoursesComponent > ngOnInit

```
11 export class CoursesComponent implements OnInit{
12     coursesService = inject(CourseService);
13     AllCourses: Course[];
14     searchString: string;
15
16     activeRoute: ActivatedRoute = inject(ActivatedRoute);
17
18     ngOnInit(){
19         // this.searchString = this.activeRoute.snapshot.queryParams['search'];
20         this.searchString = this.activeRoute.snapshot.queryParamMap.get('search');
21         console.log(this.searchString);
22
23         if(this.searchString === undefined || this.searchString === ''){
24             this.AllCourses = this.coursesService.courses;
25         }else{
26             this.AllCourses = this.coursesService.courses
27                 .filter(x => x.title.toLowerCase()
28                     .includes(this.searchString.toLowerCase()));
29         }
30     }
31 }
32
```



EXPLORER

TS app.module.ts

TS courses.component.ts

ANGULAR-ROUTING

node_modules

src

app

about

contact

courses

course-detail

courses.component.css

<> courses.component.html

TS courses.component.ts

> footer

> header

> home

> login

> Models

> not-found

Services

TS course.service.ts

TS courses.service.ts

> OUTLINE

> TIMELINE

src > app > courses > TS courses.component.ts > CoursesComponent > ngOnInit

```
11 export class CoursesComponent implements OnInit{
12     coursesService = inject(CourseService);
13     AllCourses: Course[];
14     searchString: string;
15
16     activeRoute: ActivatedRoute = inject(ActivatedRoute);
17
18     ngOnInit(){
19         // this.searchString = this.activeRoute.snapshot.queryParams['search'];
20         this.searchString = this.activeRoute.snapshot.queryParamMap.get('search');
21         console.log(this.searchString);
22
23         if(this.searchString === undefined || this.searchString === '' || this.searchString === null){
24             this.AllCourses = this.coursesService.courses;
25         }else{
26             this.AllCourses = this.coursesService.courses
27                 .filter(x => x.title.toLowerCase()
28                     .includes(this.searchString.toLowerCase()));
29         }
30     }
31 }
32
```

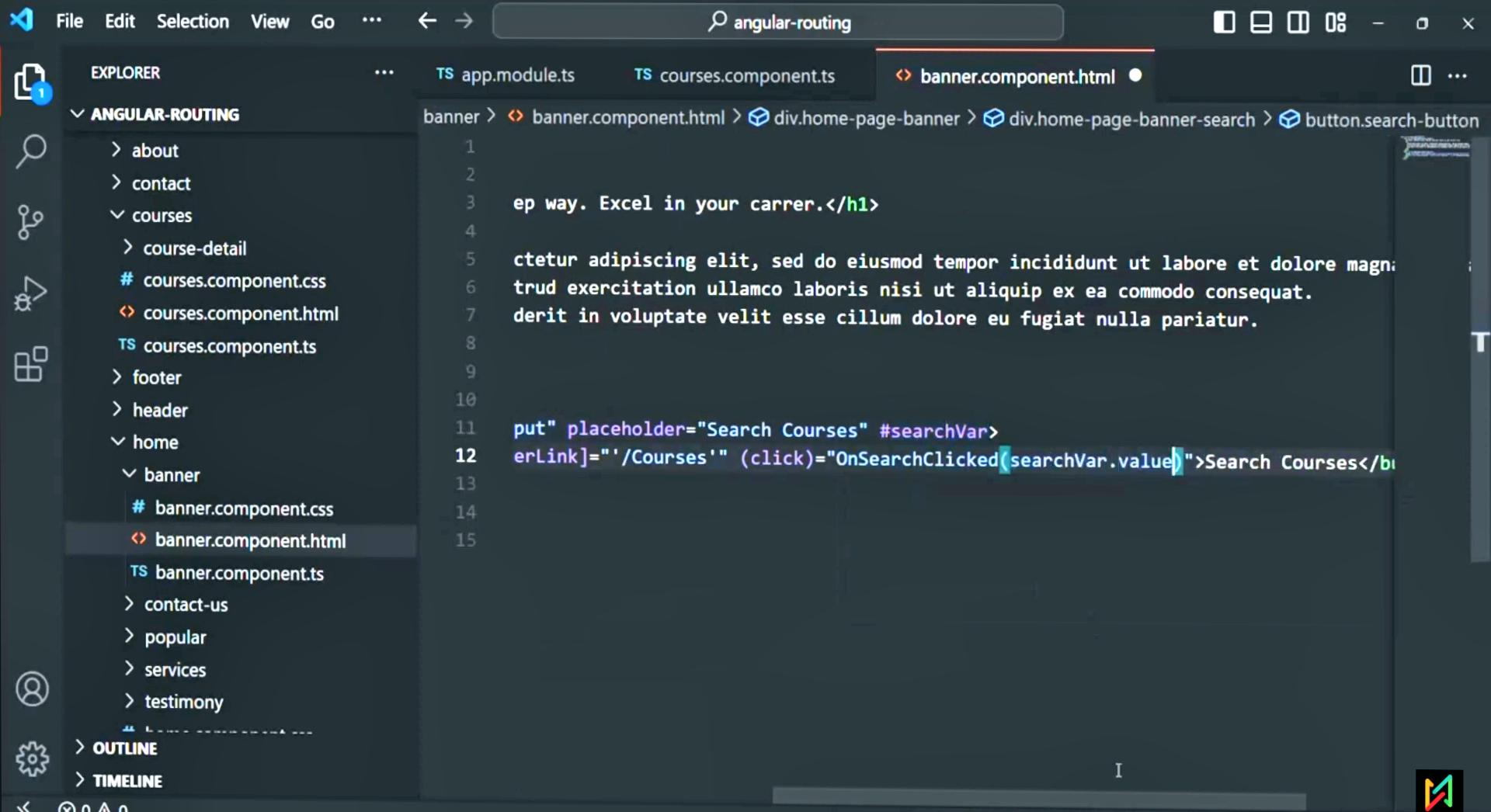
`<> banner.component.html` ●

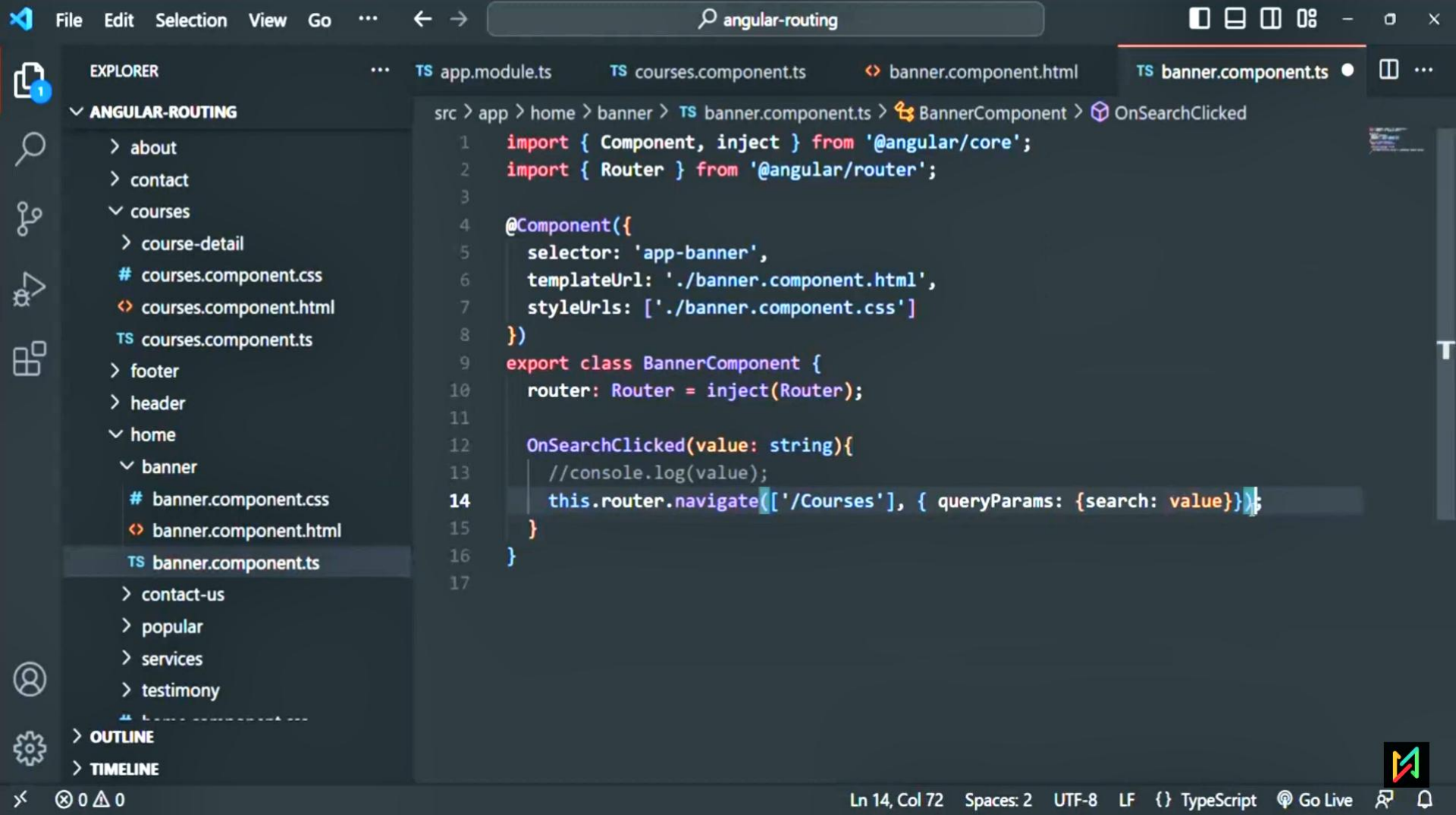
> TIMELINE

```

1  in">
2
3  step way. Excel in your carrer.</h1>
4
5  insectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore m
6  nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.
7  henderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur.
8
9
10 h">
11 <input placeholder="Search Courses" #searchVar>
12 <outerLink=""/Courses'' [queryParams]="{search: searchVar.value}">Search Courses<
13
14
15

```





EXPLORER

ANGULAR-ROUTING

- > about
- > contact
- > courses
 - > course-detail
 - # courses.component.css
 - <> courses.component.html
 - TS courses.component.ts
- > footer
- > header
- > home
 - > banner
 - # banner.component.css
 - <> banner.component.html
 - TS banner.component.ts
 - > contact-us
 - > popular
 - > services
 - > testimony

> OUTLINE

> TIMELINE

TS app.module.ts

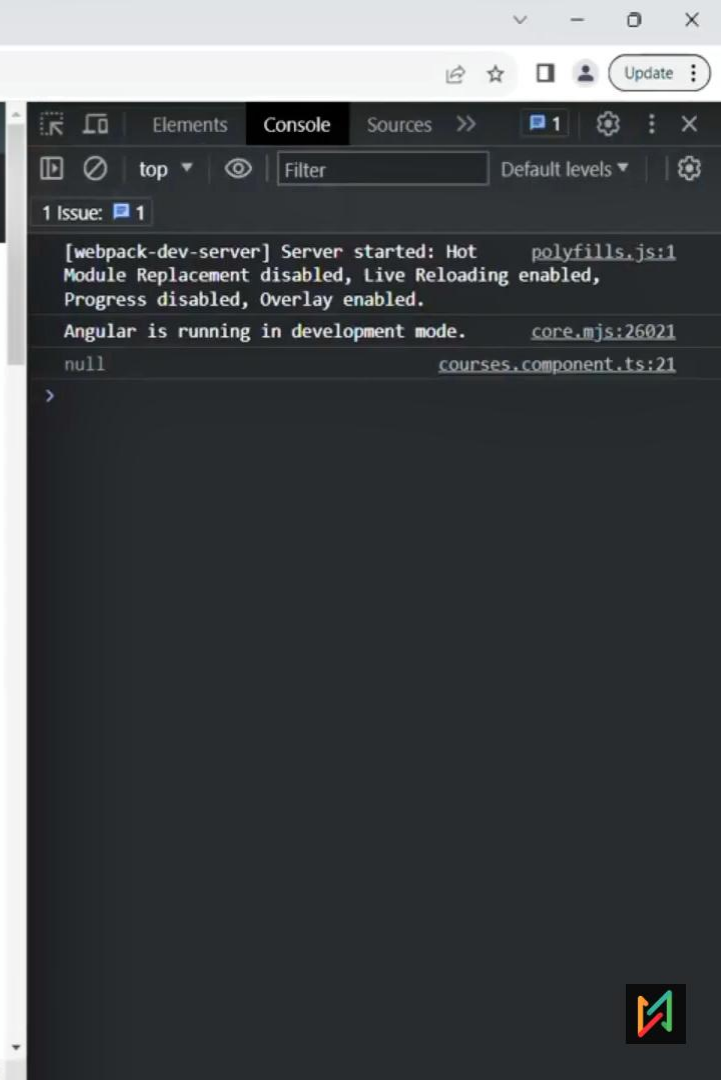
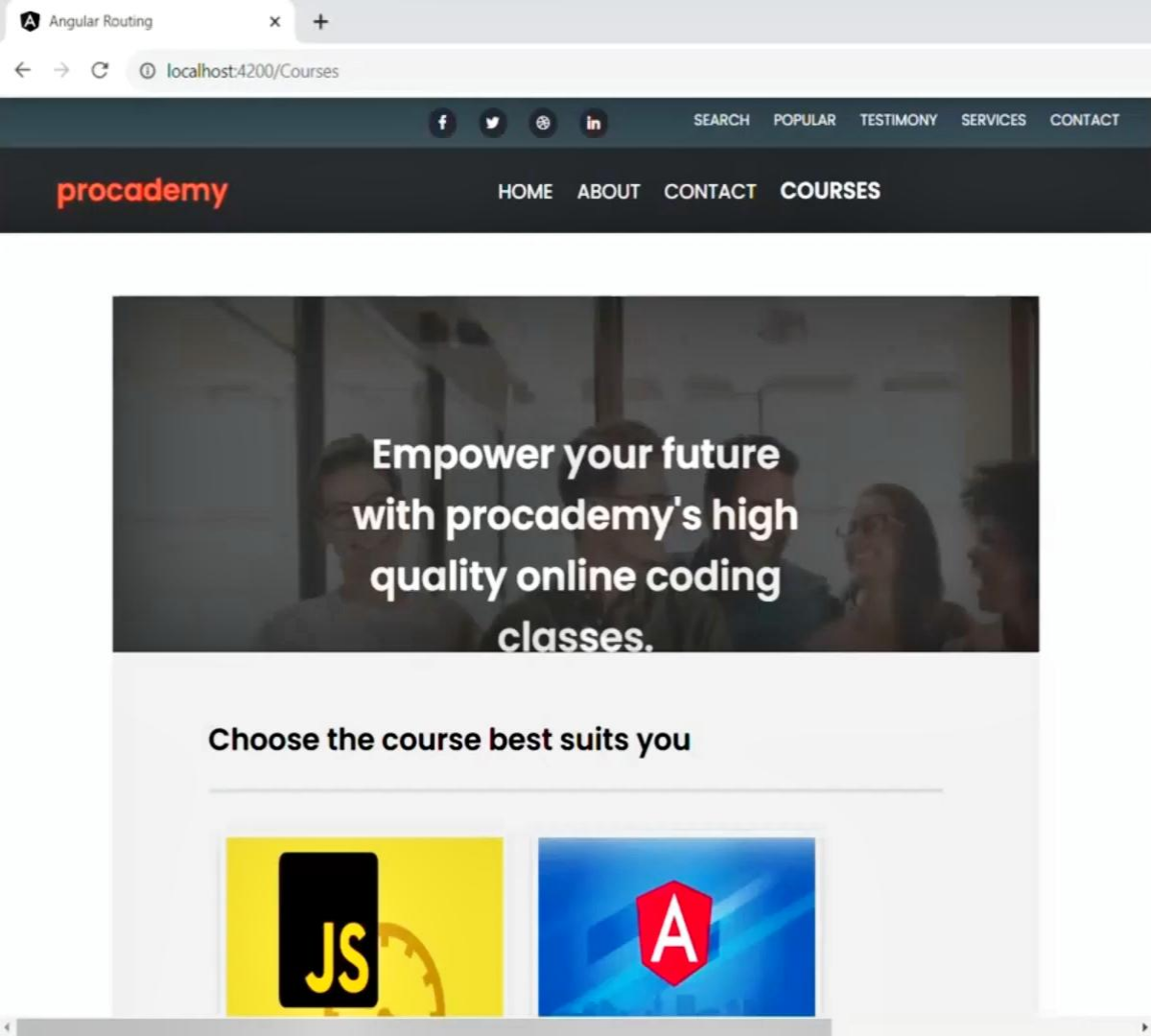
TS courses.component.ts

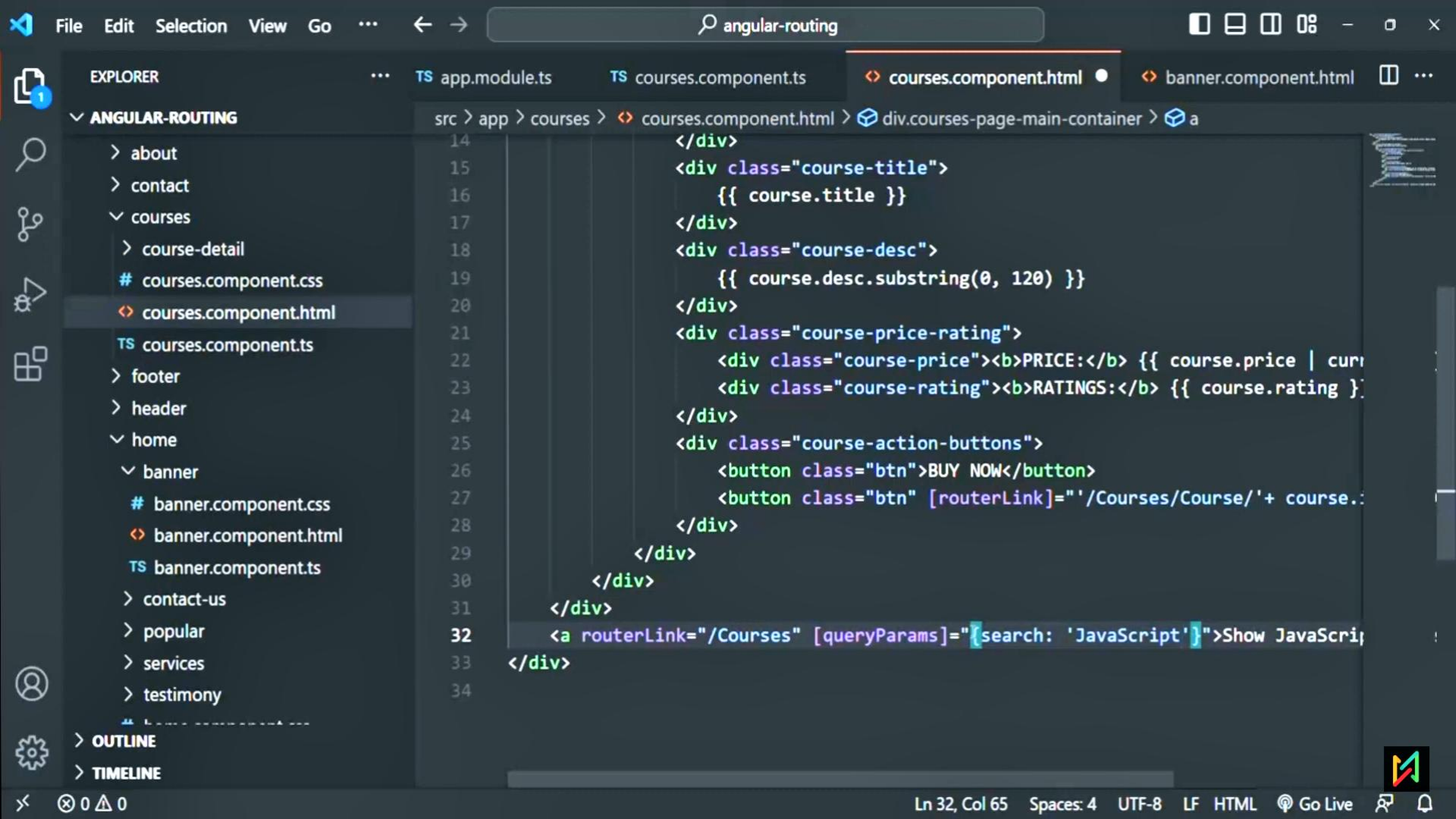
<> banner.component.html

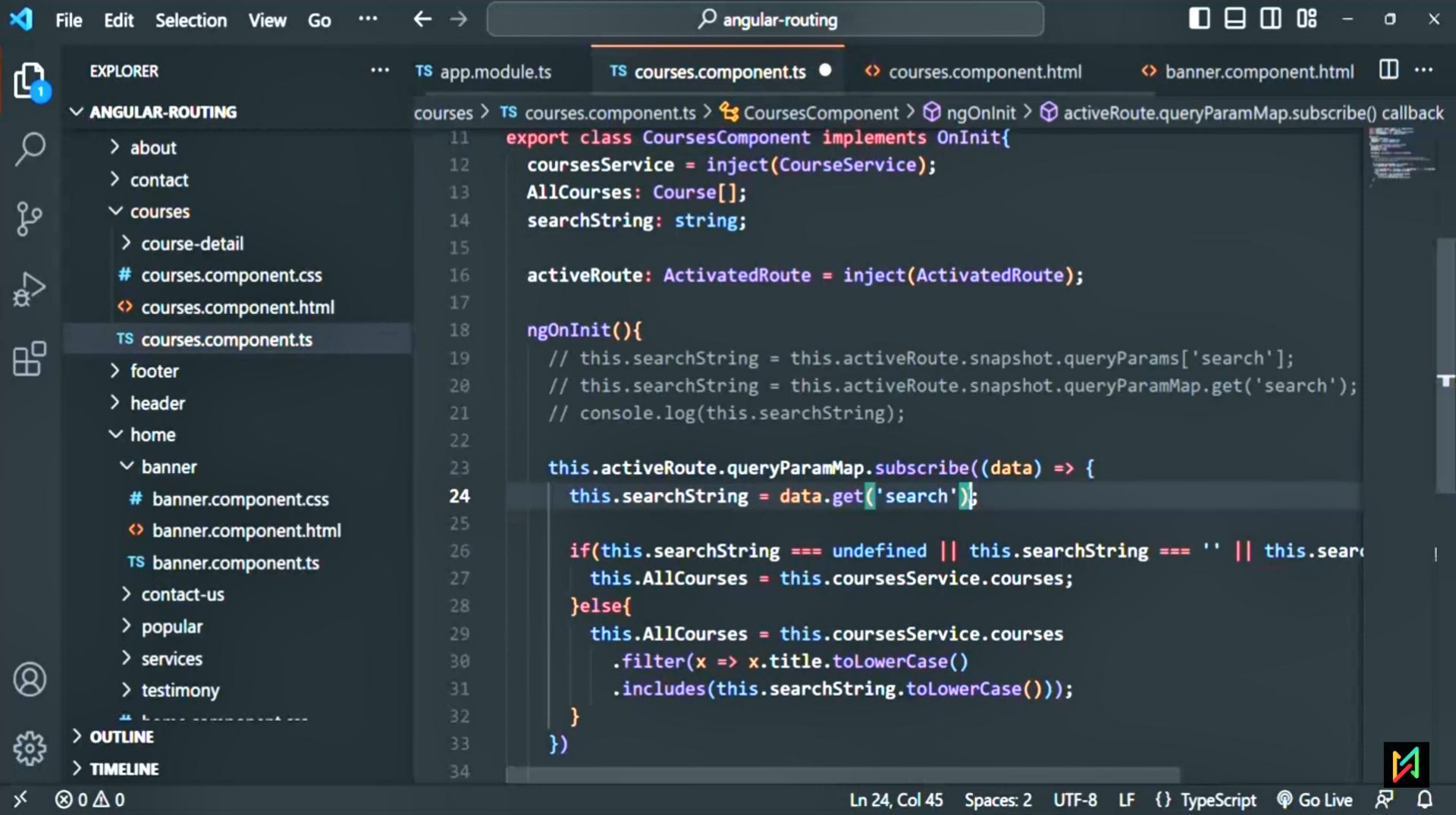
TS banner.component.ts

src > app > home > banner > TS banner.component.ts > BannerComponent > OnSearchClicked

```
1 import { Component, inject } from '@angular/core';
2 import { Router } from '@angular/router';
3
4 @Component({
5   selector: 'app-banner',
6   templateUrl: './banner.component.html',
7   styleUrls: ['./banner.component.css']
8 })
9 export class BannerComponent {
10   router: Router = inject(Router);
11
12   OnSearchClicked(value: string){
13     //console.log(value);
14     this.router.navigate(['/Courses'], { queryParams: {search: value}});
15   }
16 }
17
```





Route Parameter vs Query String

Route parameters are required and it is used by Angular router to determine the route. Route Parameters are part of route definition.

Query strings are optional data which we pass through route. If the query parameter is missing in the URL, then it will not stop angular from navigating to the route.



What is Query String

A fragment in a route is a link which jumps to a section or a content in the HTML page, which contains the ID mentioned in the fragment. A fragment comes after a # sign

`localhost:4200/Home#Services`



angular-routing

angular-routing

EXPLORED

ANGULAR-ROUTING

> header

> home

> banner

> contact-us

> popular

> services

> testimony

home.component.css

<> home.component.html

TS home.component.ts

> login

> Models

> not-found

> Services

app.component.css

<> app.component.html

TS app.component.spec.ts

TS app.component.ts

> OUTLINE

> TIMELINE

TS app.module.ts

home.component.html

src > app > home > home.component.html > div.home-page-container > section#services > app-services

1 <div class="home-page-container">

2 <section id="search">

3 | <app-banner></app-banner>

4 </section>

5

6 <section id="courses">

7 | <app-popular></app-popular>

8 </section>

9

10 <section id="testimony">

11 | <app-testimony></app-testimony>

12 </section>

13

14 <section id="services">

15 | <app-services></app-services>

16 </section>

17

18 <section class="contact">

19 | <app-contact-us></app-contact-us>

20 </section>

21 </div>

In 19, Col 22 (13 selected)

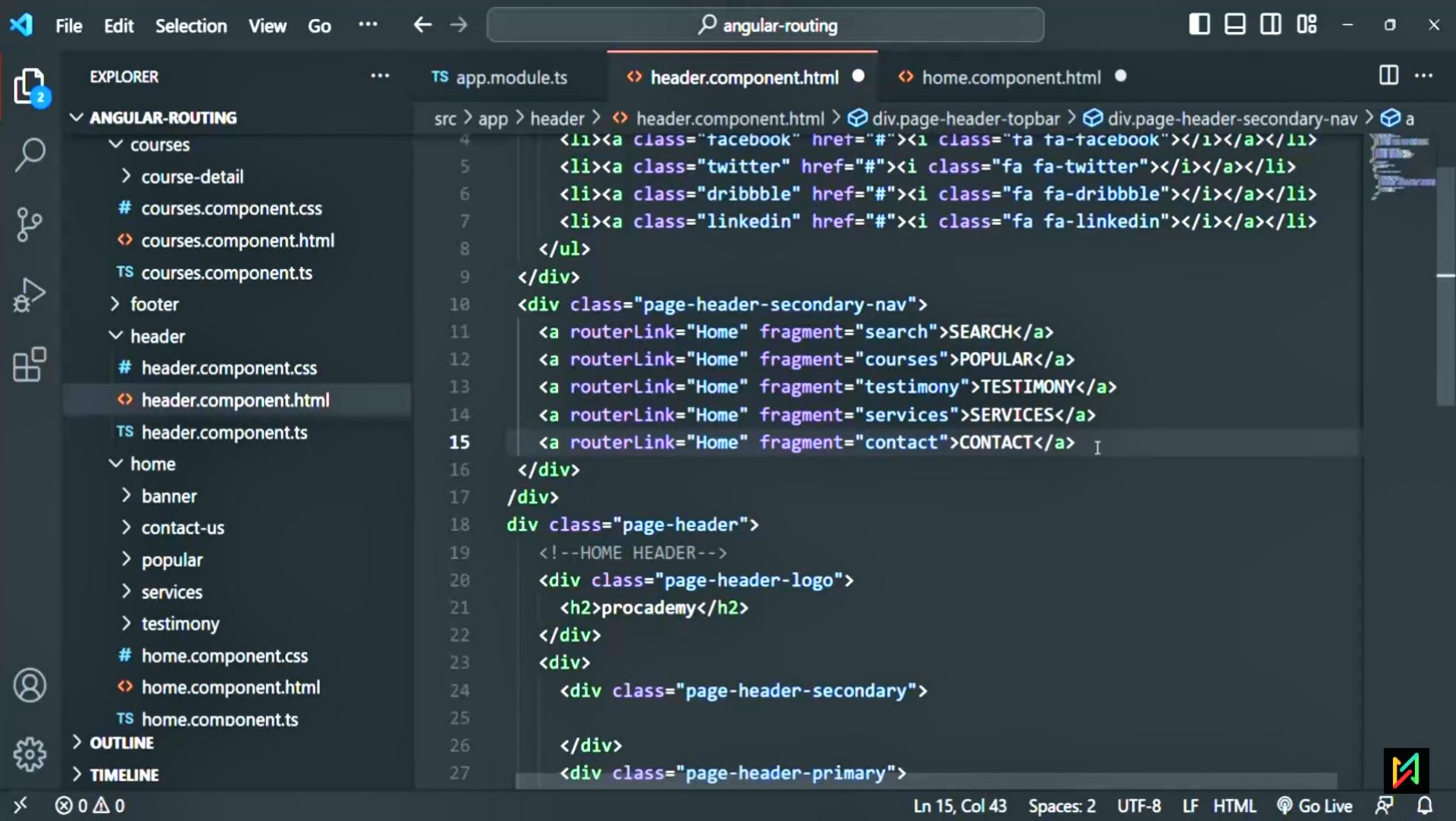
Spaces: 2

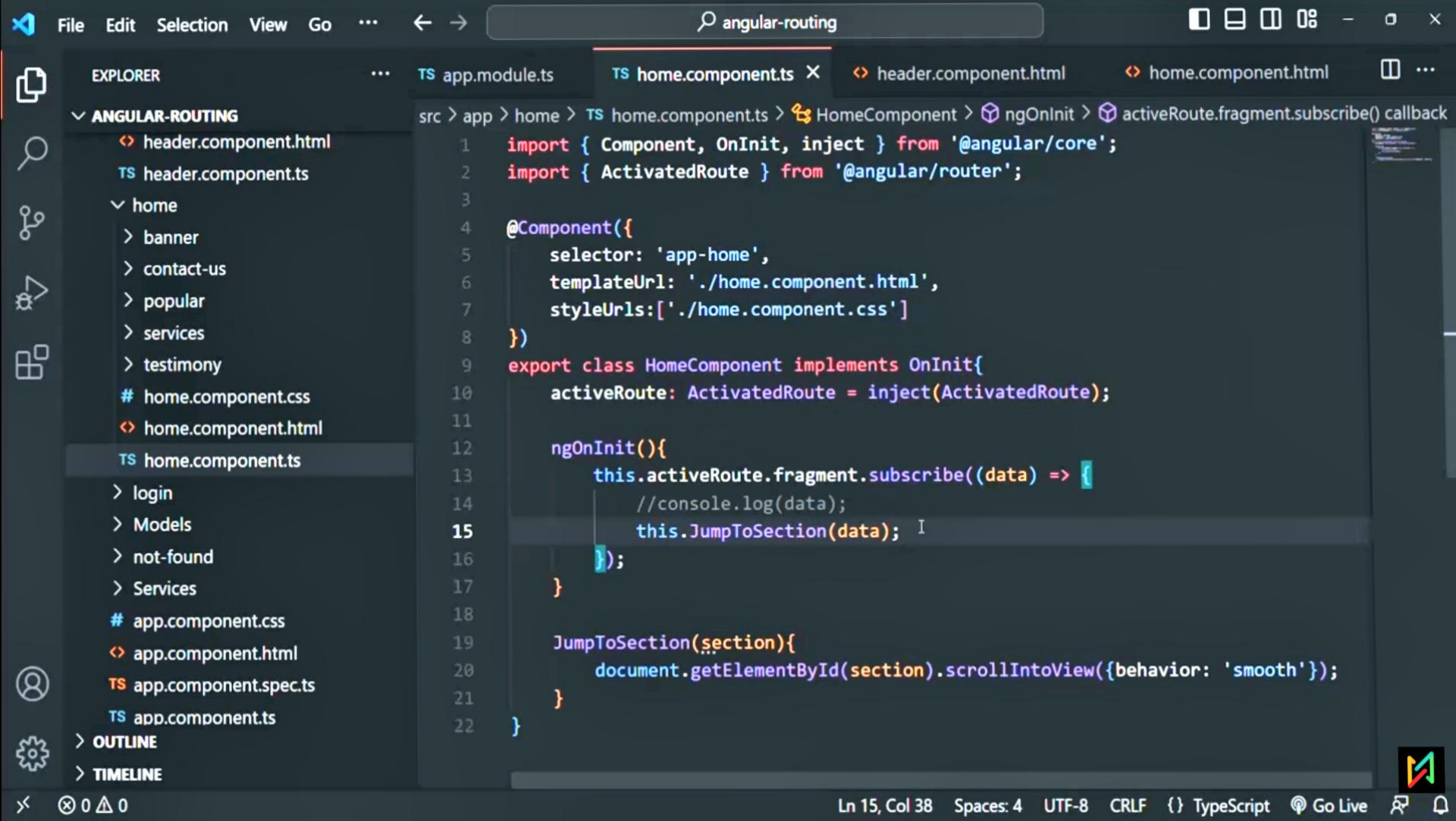
UTF-8

CRLF

HTML

Go Live





EXPLORER

ANGULAR-ROUTING

< header.component.html

TS header.component.ts

home

> banner

> contact-us

> popular

> services

> testimony

home.component.css

< home.component.html

TS home.component.ts

> login

> Models

> not-found

> Services

app.component.css

< app.component.html

TS app.component.spec.ts

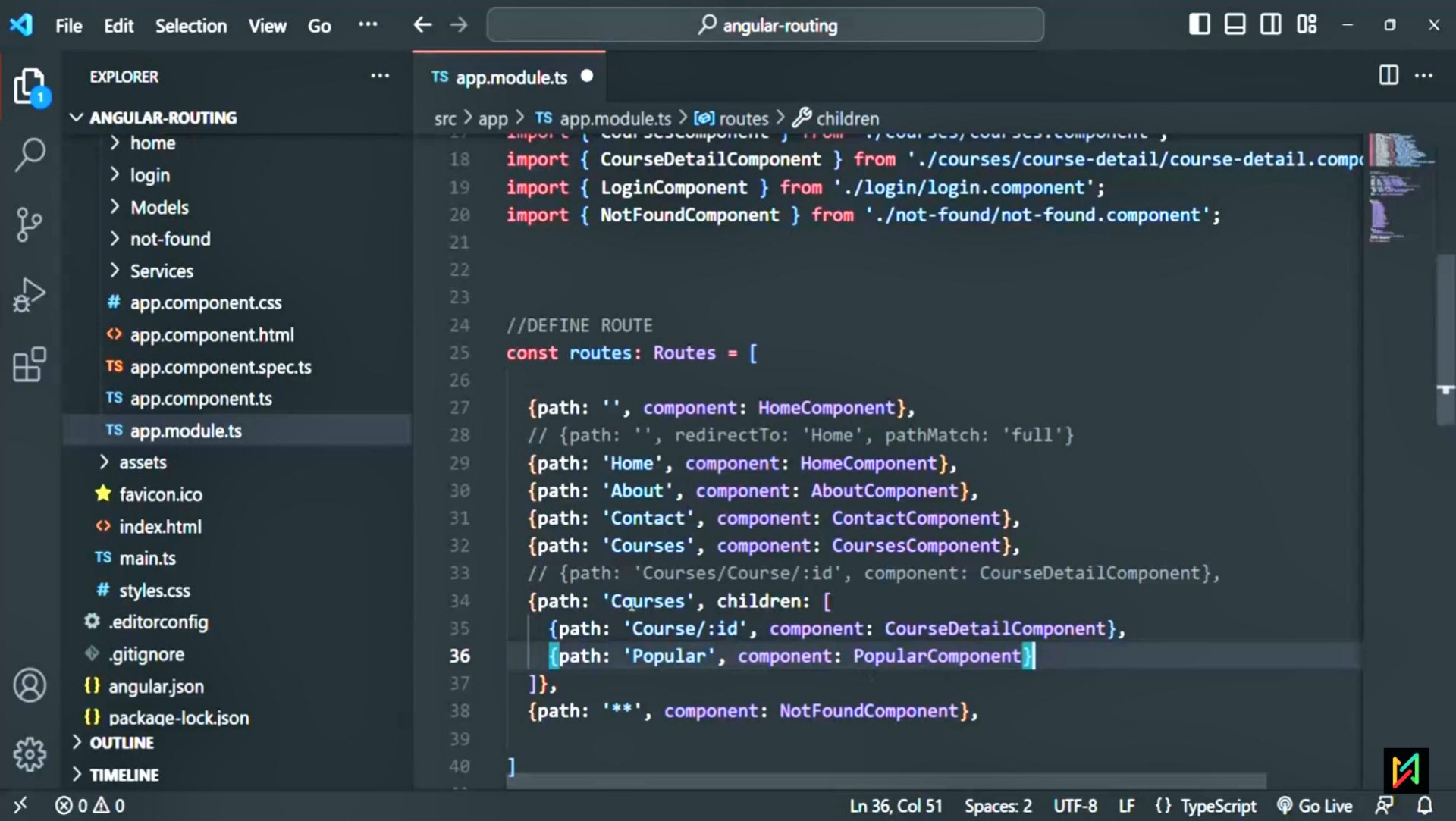
TS app.component.ts

> OUTLINE

> TIMELINE

src > app > home > TS home.component.ts > HomeComponent > ngOnInit > activeRoute.fragment.subscribe() callback

```
1  import { Component, OnInit, inject } from '@angular/core';
2  import { ActivatedRoute } from '@angular/router';
3
4  @Component({
5    selector: 'app-home',
6    templateUrl: './home.component.html',
7    styleUrls: ['./home.component.css']
8  })
9  export class HomeComponent implements OnInit {
10    activeRoute: ActivatedRoute = inject(ActivatedRoute);
11
12    ngOnInit(){
13      this.activeRoute.fragment.subscribe((data) => {
14        //console.log(data);
15        this.JumpToSection(data);
16      });
17    }
18
19    JumpToSection(section){
20      document.getElementById(section).scrollIntoView({behavior: 'smooth'});
21    }
22  }
```

EXPLORER

ANGULAR-ROUTING

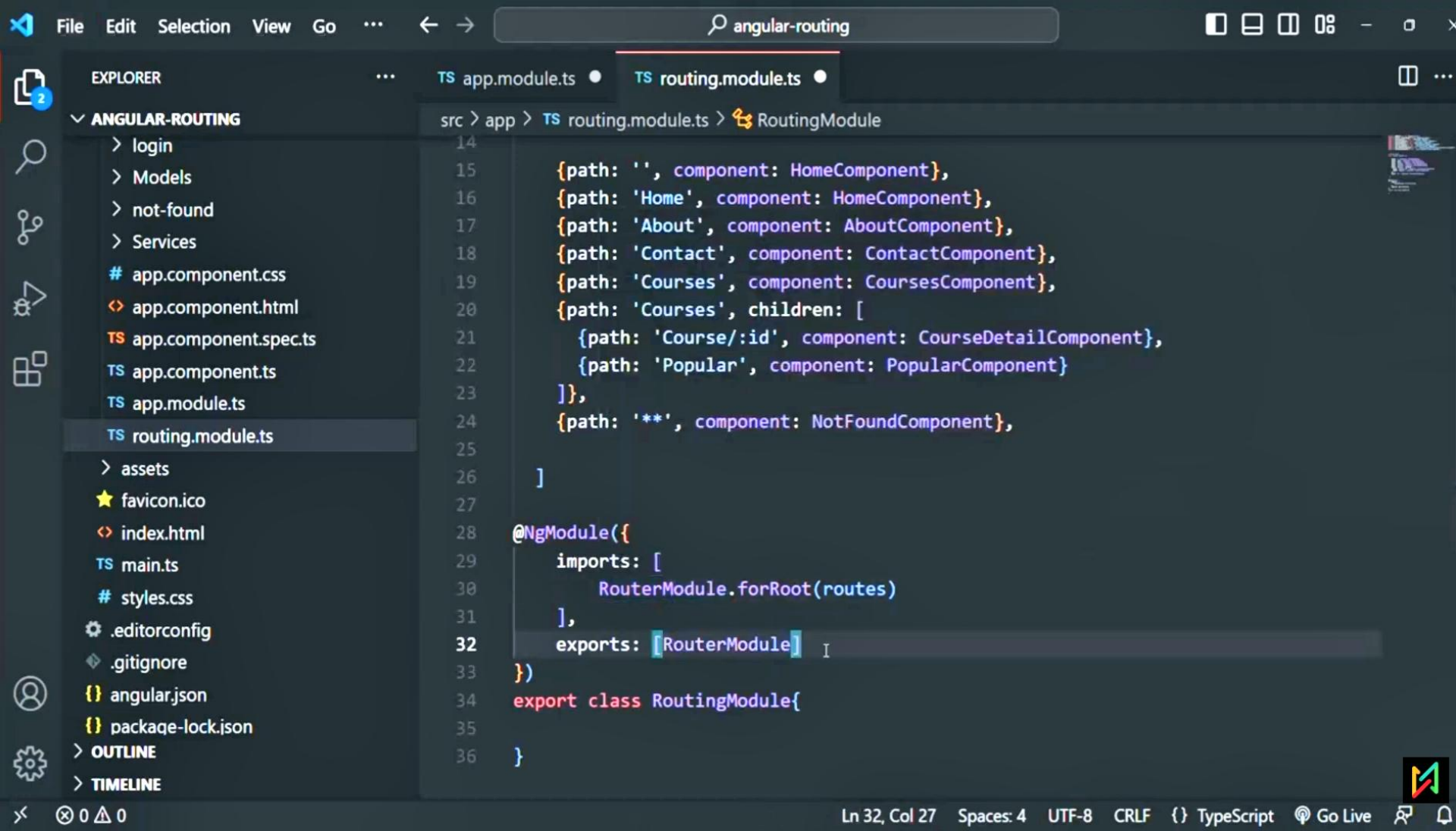
- > home
- > login
- > Models
- > not-found
- > Services
- # app.component.css
- <> app.component.html
- TS app.component.spec.ts
- TS app.component.ts
- TS app.module.ts

- > assets
- ★ favicon.ico
- <> index.html
- TS main.ts
- # styles.css
- ⚙ .editorconfig
- 🔒 .gitignore
- { } angular.json
- { } package-lock.json
- > OUTLINE
- > TIMELINE

TS app.module.ts

src > app > TS app.module.ts > routes > children

```
17 import { CoursesComponent } from './courses/courses.component';
18 import { CourseDetailComponent } from './courses/course-detail/course-detail.component';
19 import { LoginComponent } from './login/login.component';
20 import { NotFoundComponent } from './not-found/not-found.component';
21
22
23
24 //DEFINE ROUTE
25 const routes: Routes = [
26
27   {path: '', component: HomeComponent},
28   // {path: '', redirectTo: 'Home', pathMatch: 'full'}
29   {path: 'Home', component: HomeComponent},
30   {path: 'About', component: AboutComponent},
31   {path: 'Contact', component: ContactComponent},
32   {path: 'Courses', component: CoursesComponent},
33   // {path: 'Courses/Course/:id', component: CourseDetailComponent},
34   {path: 'Courses', children: [
35     {path: 'Course/:id', component: CourseDetailComponent},
36     {path: 'Popular', component: PopularComponent}
37   ]},
38   {path: '**', component: NotFoundComponent},
39
40 ]
```



EXPLORER

ANGULAR-ROUTING

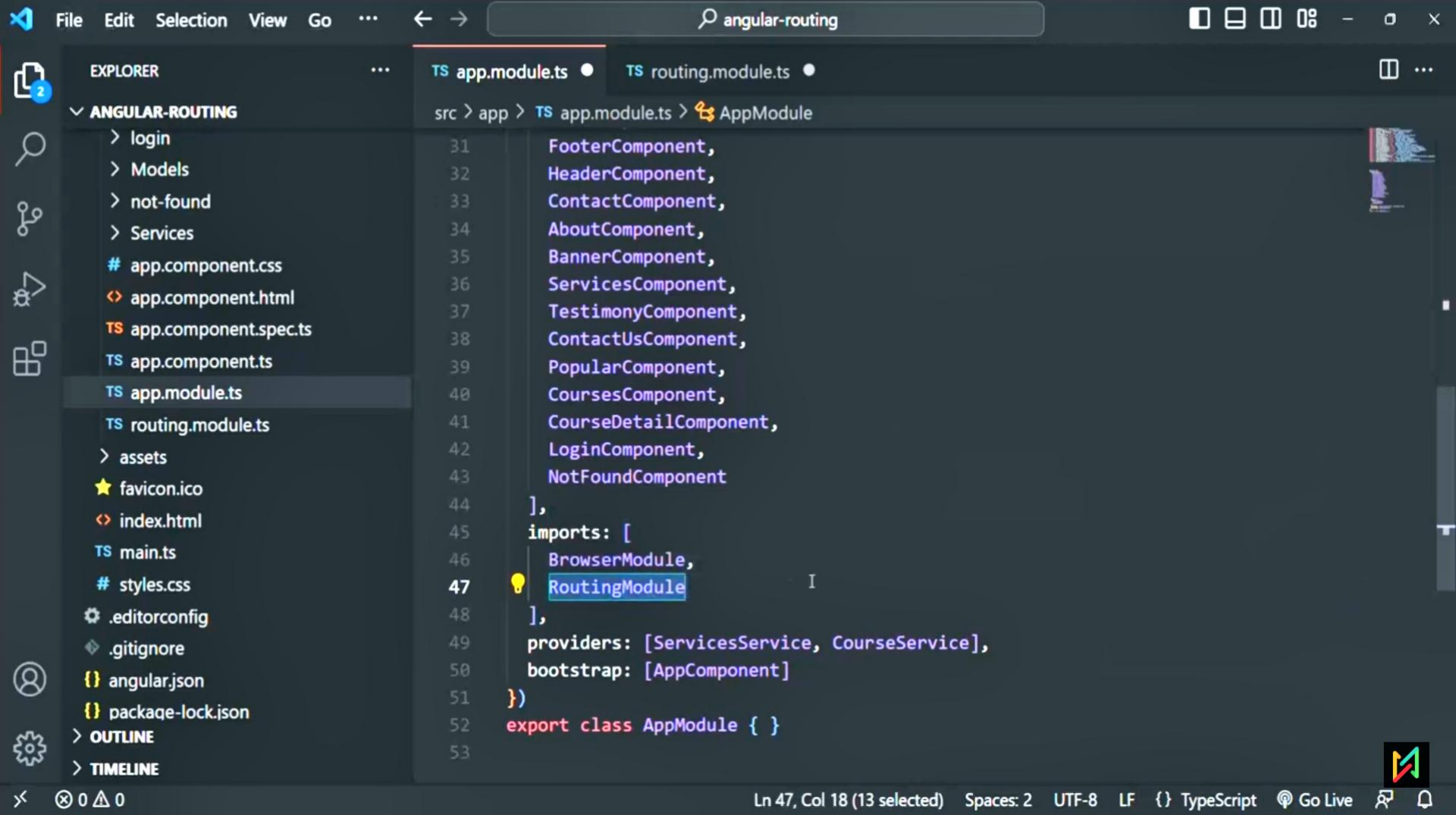
- > login
- > Models
- > not-found
- > Services
- # app.component.css
- <> app.component.html
- TS app.component.spec.ts
- TS app.component.ts
- TS app.module.ts
- TS routing.module.ts

- > assets
- ★ favicon.ico
- <> index.html
- TS main.ts
- # styles.css
- ⚙ .editorconfig
- 📄 .gitignore
- { } angular.json
- { } package-lock.json
- > OUTLINE
- > TIMELINE

TS app.module.ts • TS routing.module.ts •

src > app > TS routing.module.ts > RoutingModule

```
14
15 {path: '', component: HomeComponent},
16 {path: 'Home', component: HomeComponent},
17 {path: 'About', component: AboutComponent},
18 {path: 'Contact', component: ContactComponent},
19 {path: 'Courses', component: CoursesComponent},
20 {path: 'Courses', children: [
21   {path: 'Course/:id', component: CourseDetailComponent},
22   {path: 'Popular', component: PopularComponent}
23 ]},
24 {path: '**', component: NotFoundComponent},
25
26 ]
27
28 @NgModule({
29   imports: [
30     RouterModule.forRoot(routes)
31   ],
32   exports: [RouterModule]
33 })
34 export class RoutingModule{
35
36 }
```

Form Array



The `FormArray` is a way to manage the collection of Form Controls in Angular. The controls can be a `FormGroup`, `FormControl`, or another `FormArray`.



Form Array

We can group Form Controls in Angular forms in two ways:

1. Using FormGroup and
2. Using FormArray.

The difference is how they implement it. In FormGroup controls becomes a property of the FormGroup. Each control is represented as key-value pair.

While in FormArray, the controls become part of an array.

